

Ernest van Wijland

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Education

- **PhD Computer Science** **IRIF, Université Paris-Cité**
Paris, France *2024 – Present*
 - PhD student under Claire Mathieu (CNRS) and Vincent Cohen-Addad's (Google Research) supervision.
- **MSc Computer Science, BSc Mathematics** **École Normale Supérieure**
Paris, France *2020 – 2024*
 - Graduated both the MSc (MPRI) and the BSc (double major in Mathematics and Computer Science) with highest honors.
- **Preparatory Class** **Lycée Louis-le-Grand**
Paris, France *2018 – 2020*

Publications

- **TCS-Bench: Benchmarking State-of-the-Art Generative AI TCS Research Ability** **In Submission**
*Cohen-Addad *, Paparas *, Van Wijland *, Springer, Canitrot-Paradis, Lin, Woodruff, Kumarappan, Jayaram, Das, Jain, Svensson, Lattanzi, Balunovic, Weber, Mirrokni (*: co-first authors)* *2026*
- **Beyond Worst-Case Online Multi-Level Aggregation** **In Submission**
Mathieu, Naor, Van Wijland *2026*
- **Spectral Dual Fitting for k -Means** **In Submission**
Anand, Charikar, Cohen-Addad, Gao, Grandoni, Lee, Sharma, Van Wijland *2026*
- **A $(4 + \varepsilon)$ -Approximation for Euclidean k -Means via Non-Monotone Dual Fitting** **STOC**
Charikar, Cohen-Addad, Gao, Grandoni, Lee, Van Wijland *2026*
- **An Improved Greedy Approximation for (Metric) k -Means** **FOCS**
Charikar, Cohen-Addad, Gao, Grandoni, Lee, Van Wijland *2025*
- **Faster Approximation Scheme for Euclidean k -TSP** **SOCG**
Van Wijland, Zhou *2024*
- **Maximizing Nash Social Welfare in 2-Value Instances: Delineating Tractability** **MOOR Journal**
Akrami, Chaudhury, Hoefer, Mehlhorn, Schmalhofer, Shahkarami, Varricchio, Vermande, Van Wijland *2023*
- **Maximizing Nash Social Welfare in 2-Value Instances** **AAAI**
Akrami, Chaudhury, Hoefer, Mehlhorn, Schmalhofer, Shahkarami, Varricchio, Vermande, Van Wijland *2022*

Experience

- **Student Researcher** **Google Research**
Paris *2026*

Worked for six months under Silvio Lattanzi and Vincent Cohen-Addad's supervision at Google Research on AI's mathematical abilities.
- **PhD Student: Clustering and Network Design** **IRIF (Université Paris-Cité)**
Paris, France *2024 – Present*

Currently working under Claire Mathieu and Vincent Cohen-Addad's supervision at IRIF on finding

approximation algorithms for clustering and network design problems.

Research Intern: Approximating TSP generalizations

IRIF (Université Paris-Cité)

Paris, France

2023 – 2024

Worked for a year under Claire Mathieu and Hang Zhou's supervision at IRIF on finding fast approximation schemes for generalizations of the TSP, in Euclidean spaces and general metric spaces.

Research Intern: Euclidean CVRP, k -TSP

LIX (Polytechnique)

Palaiseau, France

2023

Worked for a semester under Hang Zhou's supervision in Polytechnique on finding faster approximation schemes for the Euclidean CVRP and k -TSP.

Research Intern: Approximating 2-WECS

THL (EPFL)

Lausanne, Switzerland

2022

Worked for a semester under Ola Svensson's supervision in the Combinatorial Optimization team at EPFL on finding a better than 2 approximation for 2-WECS.

Research Intern: Fair Division for additive instances

AGT (Max Planck Institut)

Sarrebruck, Germany

2021

Worked for two months under Kurt Mehlhorn's supervision in the Algorithmic Game Theory team of the MPII on the Fair Division problem for additive instances.

Research Intern: Spectrum of heavy-tailed random matrices

DMA (École Normale Supérieure)

Paris, France

2021

Worked under Laure Dumaz' supervision on a paper by Ben Arous and Guionnet.

Academic achievements

Best French Developer Contest 5th place finisher

Paris, France

2023

- French national contest in Computer Science, with thousands of participants from every major IT company
- awarded the 5000€ Genius prize as a high scoring student

International Olympiad in Informatics Bronze medalist

Tsukuba, Japan

2018

Dutch National Olympiad in Informatics winner

Utrecht, Netherlands

2018

Innopolis University Olympiad in Informatics finalist

Innopolis, Russia

2018

International Olympiad in Informatics participant

Tehran, Iran

2017

As part of the top 4 High-School students in the Dutch National Olympiad in Informatics.

Outreach

Member of the France-IOI association

France

2018 – Present

I am involved in organizing the Castor and Algoréa contests, in which more than 700 000 students from elementary school to high-school participate every year. I am also involved in selecting and training the best students that participate in the Informatics Olympiad.

Member of the Netherlands-IO association

Netherlands

2018 – Present

I am involved with the computer science outreach campaign towards Dutch high-school girls and preparing the Dutch girls team for the European Girls Olympiad in Informatics.

Lecturer at the Moroccan Olympiad in Informatics training camp

Morocco

2019

Gave algorithms lectures to Moroccan middle-school and high-school students.

Other interests

- Active Judo black belt and Tennis player.
- Practicing piano for the past 20 years, part of the chamber music club in the ENS. Founder of the Early Music section.
- Graduated from the Juvisy Music Conservatory in Music Theory, Piano and Ensemble Playing (2018).
- Graduated ENS with a Musicology Minor.